CO7046: RESEARCH METHODS

Week 1 — Detailed Learning Summary

Module Introduction, Understanding Research, Research Approaches and Research Types

Prepared from the Week 1 lecture slides (35 pages).

Document Overview:

This is an easy-to-revise notes from the **Day-1 Lecture: CO7046 RESEARCH METHODS** . It follows the concepts and terminology used in the slides, adds detailed explanations and examples, and concludes with important revision questions.

- First read the quick overview to understand the whole lecture.
- Study the research approaches and research types tables carefully; they are easy to confuse.
- Use the glossary and self-test questions to check your understanding.
- For official deadlines, marking criteria, reading lists and submission instructions, check the CO7046 Moodle page. The slides direct students to Moodle for resources.

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1. Week 1 at a glance

The lecture introduces CO7046 Research Methods and explains why postgraduate students need to understand how research is planned, conducted, evaluated and communicated. It then defines research, discusses the skills of a good researcher, and introduces deductive and inductive approaches plus eight common research types.

Area	Main idea
Purpose of the module	Build the skills needed to understand existing research and plan a suitable research study.
What research means	A systematic, organised, evidence-based inquiry used to answer questions or address problems.
Researcher skills	Critical thinking, analysis, curiosity, ethics, organisation, communication, literature review, methodology, data handling and attention to detail.
Research approaches	Deductive: theory → data/testing. Inductive: observations/data → patterns/theory.
Research types	Basic, applied, descriptive, analytical, qualitative, quantitative, exploratory and experimental.
Assessment	A group critical review presentation and an individual literature review/research-planning report.

2. What are CO7046 Research Methods?

The module develops the ability to investigate a topic, understand what other researchers have found, evaluate evidence critically and use relevant literature to support new insight or future research.

Topics covered in the module

- Formulating hypotheses and research questions.
- Ethical issues in science, technology and engineering research, including intellectual property and plagiarism.
- Research writing: academic style, presentation rules and referencing standards.
- Time management, library skills and literature searching.
- Choosing a research area and evaluating source material.
- Quantitative and qualitative research methods.

3. Module learning outcomes

The slides list five learning outcomes. In plain English, they describe what students should be able to do by the end of the module.

LO	Outcome from the slides, explained simply	Assessment
LO1	Choose a topic relevant to your programme, investigate it and suggest possible new directions for research.	Assessment 1
LO2	Understand what postgraduate research demands and critically evaluate current research problems.	Assessment 1
LO3	Use current research papers and other relevant resources to support a master's-level research report.	Assessment 2
LO4	Show systematic understanding and critical awareness of current practices in science, technology and engineering research.	Assessment 2
LO5	Work independently or with a group to conduct research or analyse existing research, then use findings to suggest future studies.	Assessment 2

Memory aid: LO1–LO2 = review and compare research papers. LO3–LO5 = literature review and research planning.

4. How CO7046 connects to CO7047 Research Project

- CO7046 Research Methods teaches the research skills and competence needed to carry out a project.
- CO7047 Research Project is a larger master's-level project and, according to the slides, is worth credits equivalent to three taught modules.
- The project assessment described in the slides is a 10,000-word report plus a poster presentation.
- The slides state that an original contribution to knowledge is not required in the way it is for a PhD. Replicating published work can be an option, but the student's written report must be original.
- Passing the Research Project is required for the MSc qualification.

Practical meaning: use CO7046 to learn how to find credible research, assess it, choose suitable methods and plan a manageable project before undertaking CO7047.

5. Assessment overview

Assessment	What you do	Format stated in slides	Learning outcomes
1. Critical Review	Critically review and compare two research papers supplied by the module tutor.	30-minute group presentation; submit presentation slides through Turnitin.	LO1 and LO2
2. Literature Review and Research Planning	Review literature on a current topic you choose that relates to Computer Science, Cybersecurity or Engineering. The topic may be suitable for CO7047.	1,700-word report submitted through Turnitin.	LO3, LO4 and LO5

- The pass mark stated in the slides is 50%.
- The slides warn that retake opportunities are limited and retake marks are capped at 50%.
- Always check the current Moodle brief for the exact deadline, marking rubric, format, citation style and any updated instructions; these details are not fully specified in the slides.

6. Academic research: evidence, perspective and assumptions

The lecture uses several examples to encourage students to question assumptions and avoid treating one limited perspective as the complete truth.

“Standing on the shoulders of giants”

The slides refer to Isaac Newton's well-known expression. In research, it represents how new work builds on the discoveries, methods and ideas of earlier researchers. Reading and crediting earlier work helps a researcher understand what is already known and identify what might be investigated next.

The story of the blind men and the elephant

Each person touches a different part of an elephant and concludes that the whole animal resembles a wall, spear, rope, tree, snake or fan. Each observation contains part of the truth, but none is sufficient to describe the whole elephant.

Research lesson: an observation or study may provide only a partial view. Researchers should consider evidence from different sources, understand the limits of a method, compare explanations and avoid overgeneralising from limited data.

The Chernobyl example and the importance of truth

The slides use the Chernobyl disaster and the line “Every lie we tell incurs a debt to the truth. Sooner or later, that debt is paid” to emphasise that truth and transparency matter. In academic work, this connects to reporting evidence honestly, not hiding inconvenient results, and not misrepresenting sources or findings.

What this means for a cybersecurity researcher

- Do not assume one alert, log entry or dataset tells the whole story of an incident.
- Compare evidence from multiple relevant sources where possible.
- Separate what the evidence shows from what you suspect it means.
- State limitations and uncertainty rather than presenting a guess as a proven fact.
- Give credit to other people's ideas, words, code and research.

7. What is research?

The lecture defines research as a systematic and organised process of inquiry intended to generate information that answers questions or addresses problems. It involves structured collection and analysis of data, reasoning and evidence-based methods, and aims to produce ethical, reliable and valid conclusions.

Key words in simple language

Term	Plain-language meaning
Systematic	Follow a planned, logical process rather than working randomly.
Inquiry	A careful investigation into a question or problem.
Data	Information collected for the study, such as measurements, logs, survey responses or interview answers.
Evidence-based	Base conclusions on information that supports them, not just opinion.
Ethical	Conduct the study responsibly and follow relevant rules and protections.
Reliable	The process or measurement is consistent enough to produce dependable results.
Valid	The method and evidence actually address what the study claims to investigate.
Conclusion	A reasoned answer based on the evidence analysed.

Simple example: if you want to investigate whether a new intrusion-detection algorithm improves detection, define the question, choose suitable test data and evaluation measures, run a planned comparison, analyse the results and explain what the evidence does—and does not—show.

8. Skills of a good researcher

Skill	What it means	Cybersecurity-related example
Critical thinking	Question assumptions and judge the strength of evidence.	Do not label an event malicious based on one indicator alone.
Analytical skills	Interpret information, spot patterns and draw reasoned conclusions.	Compare incident records to find recurring attack patterns.
Adaptability and curiosity	Stay open to new ideas and revise an approach when findings require it.	Change the research question or method if initial evidence reveals a different issue.
Ethical awareness	Understand and follow ethical standards.	Protect personal data and obtain appropriate permission for research involving people or systems.
Problem-solving	Define a problem clearly and identify a suitable way to investigate it.	Turn “network security is poor” into a focused, answerable research question.
Time management and organisation	Plan tasks, manage sources and meet milestones.	Schedule searching, reading, analysis and writing before the deadline.
Literature review skills	Find, evaluate and combine relevant academic sources.	Compare several papers on phishing detection rather than summarising only one.
Research design and methodology	Choose methods that can answer the research question reliably and validly.	Select a controlled experiment if comparing algorithm performance under the same conditions.
Data collection and analysis	Collect information appropriately and analyse it using suitable techniques.	Evaluate detection rates, false positives and other chosen metrics.
Communication skills	Explain methods, evidence and conclusions clearly in reports and presentations.	Describe how an experiment was conducted so another researcher can understand it.
Attention to detail	Maintain accuracy and consistency at each stage.	Check dataset labels, experiment settings, citations and calculations.

9. Planning and describing your research

Know your field

Understand common research approaches and methods in your discipline and why researchers use them. Do not select a method simply because it is familiar.

Match the method to the question

Start by deciding what you need to find out. Then choose a method that can reasonably answer that question.

Know your audience

Understand how researchers in your field communicate, what evidence they expect and what they consider persuasive.

Use the literature review to guide planning

Reviewing previous work helps reveal what is known, how it has been studied, what methods are common and where a useful question may remain.

Be aware of your own biases

Notice how your expectations or preferences might affect source selection, interpretation or conclusions.

10. Research approaches: deductive vs inductive

Feature	Deductive approach	Inductive approach
Direction	Top-down	Bottom-up
Starts with	A theory or hypothesis	Observations or collected data
Main process	Use data to test whether the theory or hypothesis is supported.	Look for patterns in data and develop a possible explanation or theory.
Often associated with	Quantitative and experimental research	Qualitative and exploratory research
Simple example	Hypothesis: algorithm A has higher detection accuracy than algorithm B. Test both on the same dataset.	Review a collection of incident reports, notice recurring behaviours and develop a possible explanation for a pattern.

Memory trick: Deductive = theory first, test it. Inductive = observations first, build an explanation. The slides say these approaches are often associated with certain research types; “often” does not mean they are exclusive to them.

11. Eight research types introduced in the lecture

The slides classify research by different dimensions: purpose (such as basic or applied), the kind of data (quantitative or qualitative), and the design or aim (such as experimental or descriptive). These labels are not all competing alternatives on one single scale. A study can fit more than one label depending on its purpose and method.

Type	Meaning	Example	Question to remember
Basic (pure) research	Builds or expands theoretical knowledge without an immediate practical application being the main goal.	Developing or refining a theory about a scientific or mathematical phenomenon.	Is the main aim to expand knowledge?
Applied research	Targets a specific real-world problem and seeks knowledge that can support a practical solution.	Developing and evaluating a cybersecurity intrusion-detection system.	Is the study aimed at solving a practical problem?
Descriptive research	Describes characteristics of a population, situation or phenomenon; it does not manipulate the variables being studied.	Summarising the types of cyber incidents reported by an organisation over a period.	What does the situation look like?
Analytical research	Uses existing information to investigate relationships, patterns or possible causes, going beyond simply describing what happened.	Analysing breach records to identify factors associated with successful attacks.	What patterns or explanations can be found in existing information?
Qualitative research	Explores experiences, behaviour, meanings or perspectives; data may be words, themes or observations.	Interviewing security staff about how they respond to alerts.	How do people experience, understand or explain something?
Quantitative research	Measures variables using numerical data and analyses them, often with statistics.	Measuring detection accuracy or the number of false positives produced by a tool.	How much, how often, or what numerical relationship?
Exploratory research	Investigates a new or poorly understood issue and helps develop questions or hypotheses for later research.	Exploring a newly observed malware type when little previous research exists.	What might be happening, and what should be studied next?
Experimental research	Manipulates an independent variable and observes its effect on a dependent variable, usually under controlled conditions.	Testing whether changing an algorithm improves network performance in a lab or simulation.	Does changing X affect Y under controlled conditions?

12. Important research terms

Term	Easy explanation	Example
Research question	The focused question the study is designed to answer.	Does algorithm A detect more phishing URLs than algorithm B on the same test set?
Hypothesis	A testable prediction or proposed relationship.	Algorithm A will achieve higher detection accuracy than algorithm B.
Independent variable	The factor deliberately changed in an experiment.	Which detection algorithm is used.
Dependent variable	The outcome measured to see whether it changes.	Detection accuracy or false-positive rate.
Controlled conditions	Keeping relevant conditions consistent so comparisons are fair.	Use the same dataset and evaluation procedure for both algorithms.
Literature review	A structured review and synthesis of relevant previous research.	Compare methods, findings and limitations across papers on phishing detection.
Methodology	The overall reasoning and plan for how the study will be carried out.	A controlled comparison using a labelled dataset and predefined metrics.
Bias	A systematic influence that may distort what is collected or how it is interpreted.	Only selecting papers that support your preferred algorithm.
Plagiarism	Presenting another person's words, ideas or work as your own without appropriate acknowledgement.	Copying a paper's explanation without quotation or citation.
Referencing	Giving readers details of sources used so the work can be traced and credited.	Citing the paper from which a method or claim came.

13. How to choose a suitable research type

Ask these questions in order. A single study may have more than one valid description, so focus on the main purpose and the method actually used.

1. Purpose: Am I mainly expanding knowledge (basic) or solving a practical problem (applied)?
2. Current state of knowledge: Is the topic poorly understood (exploratory), or am I describing a known situation (descriptive)?
3. Data: Am I primarily interpreting words, experiences or themes (qualitative), or measuring numbers (quantitative)?
4. Analysis: Am I examining existing information for patterns or explanations (analytical)?
5. Design: Am I deliberately changing a variable under controlled conditions to measure an effect (experimental)?

14. Worked examples based on the lecture's group activity

Scenario	Most appropriate answer	Why
Study the socioeconomic characteristics of residents in a country.	Descriptive	The goal is to describe characteristics of a population.
Investigate why employees feel more motivated in remote working environments.	Qualitative	The wording asks about experiences, feelings and reasons; qualitative methods can explore these in depth.
Measure employee productivity using numerical performance metrics.	Quantitative	The evidence is numerical and is measured.
Analyse existing cyberattack data to identify causes and patterns.	Analytical	The study uses existing information to investigate patterns and possible causes.
Test whether a new algorithm improves network performance under controlled conditions.	Experimental	A variable is changed and its effect is measured under controlled conditions.
Develop and evaluate a real-time cyberattack detection system to address a practical security problem.	Applied	The main aim is to address a practical security need.
Investigate a completely new type of malware where no prior studies exist.	Exploratory	The topic is new or poorly understood and requires initial investigation.

These answers follow the options and definitions shown in the lecture activity. Real projects can combine types—for example, applied research can include an experiment and quantitative evaluation.

15. Practical advice for postgraduate study

- Check your university email and Moodle regularly.
- Attend sessions on time and complete attendance check-in when required.
- Watch lecture recordings and complete the tasks set for each lesson.
- Read beyond the lecture slides: the slides are a starting point, not the entire research literature.
- Question information logically and evaluate the quality of evidence.
- Read assessment instructions carefully and start early enough to seek guidance.
- Ask tutors for help well before deadlines.
- Understand and follow rules on plagiarism and academic malpractice.
- Respect other students' opportunity to learn.

16. Quick revision sheet

If you see this phrase…	Think…
Theory or hypothesis first, then test with data	Deductive approach
Observe data first, then develop an explanation	Inductive approach
Expand knowledge without immediate practical use	Basic research
Solve a real-world problem	Applied research
Describe what a population or situation is like	Descriptive research
Use existing data to examine patterns or causes	Analytical research
Words, experiences, meanings and themes	Qualitative research
Numbers, measurements and statistics	Quantitative research
New or poorly understood topic	Exploratory research
Change a variable and measure its effect under controlled conditions	Experimental research
Assessment 1	30-minute group critical review presentation; LO1 and LO2
Assessment 2	1,700-word individual literature review and research-planning report; LO3, LO4 and LO5

17. Self-test questions

1. In your own words, define research.

2. Name three qualities of ethical, reliable research.

3. What is the difference between a research question and a hypothesis?

4. Explain the difference between deductive and inductive approaches.

5. A researcher interviews security analysts about their experiences of alert fatigue. Which research type is especially relevant, and why?

6. A researcher counts false positives from two tools and compares their results. Which data approach is central?

7. A researcher deliberately changes the algorithm while keeping test conditions consistent. What research design is suggested?

8. What is the difference between descriptive and analytical research?

9. What is the difference between basic and applied research?

10. Which learning outcomes are assessed in Assessment 1? Which are assessed in Assessment 2?

11. Why is it important to read existing literature before selecting a research method?

12. How can the blind men and the elephant story help a researcher avoid mistakes?

Suggested answers

- A planned, organised, evidence-based investigation to answer questions or address problems.
- For example: it follows a systematic process, uses evidence, follows ethical standards, and aims for reliable and valid conclusions.
- A research question asks what the study wants to find out; a hypothesis is a testable prediction.

- Deductive starts with a theory or hypothesis and tests it; inductive starts with observations/data and develops a theory or explanation.
- Qualitative research, because it explores analysts' experiences and meanings.
- Quantitative research, because false positives are numerical measurements.
- Experimental research, because a variable is deliberately changed under controlled conditions.
- Descriptive research characterises a situation; analytical research examines existing information for relationships, patterns or possible causes.
- Basic research mainly expands knowledge; applied research targets a practical problem.
- Assessment 1: LO1 and LO2. Assessment 2: LO3, LO4 and LO5.
- Literature shows what is known, what methods are used, what gaps remain and how to choose a suitable approach.
- One perspective may be partial. Researchers should compare evidence, question assumptions and avoid claiming that one limited observation proves the whole picture.

18. References and source boundaries

Primary source: CO7046: Research Methods, Week 1: Module Introduction and Overview, lecture slides supplied by the user (35 pages).

The slides direct students to the “Resource List for CO7046” on Moodle and the University of Chester Library Search for key references. The detailed bibliographic entries are not included in the extracted Week 1 slides, so this summary does not invent them.

This document is a study aid, not a replacement for the official module handbook, Moodle assessment brief or tutor guidance.
